

Q5-A: Explain the second fundamental form of a surface. Derive the expression for the normal curvature and provide a geometrical interpretation.

1 The Second Fundamental Form

1.1 Motivation

The first fundamental form tells us nothing about how the surface bends in $\mathbb{R}^3$. To capture bending, we need to compare the surface to its tangent plane by measuring how much the unit normal $\hat{\mathbf{n}}$ changes as we move along the surface.

1.2 Derivation of the Coefficients

Since $\hat{\mathbf{n}} \cdot \mathbf{r}_u = 0$ and $\hat{\mathbf{n}} \cdot \mathbf{r}_v = 0$ (the normal is perpendicular to all tangent vectors), differentiating the first equation with respect to u gives

$$\hat{\mathbf{n}}_u \cdot \mathbf{r}_u + \hat{\mathbf{n}} \cdot \mathbf{r}_{uu} = 0 \implies \hat{\mathbf{n}} \cdot \mathbf{r}_{uu} = -\hat{\mathbf{n}}_u \cdot \mathbf{r}_u.$$

This shows that the second-order variation of $\mathbf{r}$ projected onto $\hat{\mathbf{n}}$ equals (up to sign) the variation of $\hat{\mathbf{n}}$ projected back onto the tangent plane.

Definition 1 (Second Fundamental Form). The **second fundamental form** is the symmetric bilinear form

$$\text{II} = L du^2 + 2M du dv + N dv^2,$$

where the coefficients are defined by

$$L = \mathbf{r}_{uu} \cdot \hat{\mathbf{n}}, \quad M = \mathbf{r}_{uv} \cdot \hat{\mathbf{n}}, \quad N = \mathbf{r}_{vv} \cdot \hat{\mathbf{n}}.$$

Equivalently, using $|\mathbf{r}_u \times \mathbf{r}_v| = \sqrt{EG - F^2}$,

$$L = \frac{(\mathbf{r}_{uu}, \mathbf{r}_u, \mathbf{r}_v)}{\sqrt{EG - F^2}}, \quad M = \frac{(\mathbf{r}_{uv}, \mathbf{r}_u, \mathbf{r}_v)}{\sqrt{EG - F^2}}, \quad N = \frac{(\mathbf{r}_{vv}, \mathbf{r}_u, \mathbf{r}_v)}{\sqrt{EG - F^2}},$$

where $(\mathbf{a}, \mathbf{b}, \mathbf{c})$ denotes the scalar triple product $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.

Remark. Unlike E, F, G (which depend only on the metric), the coefficients L, M, N depend on the *embedding* of the surface in $\mathbb{R}^3$ and on the choice of normal orientation.

2 Derivation of Normal Curvature

2.1 Normal Sections

Definition 2 (Normal Section). Let P be a point on the surface and $\mathbf{t}$ a unit tangent vector at P . The **normal section** in the direction $\mathbf{t}$ is the curve γ obtained by intersecting the surface with the plane spanned by $\mathbf{t}$ and $\hat{\mathbf{n}}(P)$.

2.2 Step-by-Step Derivation

Let $\gamma(t) = \mathbf{r}(u(t), v(t))$ be a smooth curve on the surface passing through $P = \mathbf{r}(u_0, v_0)$ at $t = 0$, with unit speed: $|\dot{\gamma}| = 1$.

Step 1: First derivative.

$$\dot{\gamma} = \mathbf{r}_u \dot{u} + \mathbf{r}_v \dot{v}.$$

Step 2: Second derivative.

$$\ddot{\gamma} = \mathbf{r}_{uu} \dot{u}^2 + 2\mathbf{r}_{uv} \dot{u} \dot{v} + \mathbf{r}_{vv} \dot{v}^2 + \mathbf{r}_u \ddot{u} + \mathbf{r}_v \ddot{v}.$$

Step 3: Project onto the normal. Since $\mathbf{r}_u$ and $\mathbf{r}_v$ are tangent vectors, $\hat{\mathbf{n}} \cdot \mathbf{r}_u = \hat{\mathbf{n}} \cdot \mathbf{r}_v = 0$. Therefore

$$\ddot{\gamma} \cdot \hat{\mathbf{n}} = L \dot{u}^2 + 2M \dot{u} \dot{v} + N \dot{v}^2.$$

Step 4: Identify the normal curvature. For a unit-speed curve, the curvature vector is $\ddot{\gamma}$. Its component along $\hat{\mathbf{n}}$ is the **normal curvature**:

$$\kappa_n = \ddot{\gamma} \cdot \hat{\mathbf{n}} = L \dot{u}^2 + 2M \dot{u} \dot{v} + N \dot{v}^2.$$

Step 5: Express as a ratio of fundamental forms. For an arbitrary (not necessarily unit-speed) curve with tangent $d\mathbf{r} = \mathbf{r}_u du + \mathbf{r}_v dv$, we have

$$I = E du^2 + 2F du dv + G dv^2 = |d\mathbf{r}|^2 > 0,$$

so the unit-speed condition normalises the denominator to 1. In general,

Theorem 3 (Normal Curvature Formula). *The normal curvature of a surface $\mathbf{r}(u, v)$ in the tangent direction $(du : dv)$ is*

$$\kappa_n = \frac{II}{I} = \frac{L du^2 + 2M du dv + N dv^2}{E du^2 + 2F du dv + G dv^2}.$$

For a unit tangent vector $\mathbf{t} = a\mathbf{r}_u + b\mathbf{r}_v$ with $Ea^2 + 2Fab + Gb^2 = 1$, this simplifies to

$$\kappa_n = La^2 + 2Mab + Nb^2.$$

3 Principal Curvatures, Gaussian and Mean Curvature

3.1 Principal Curvatures

The normal curvature κ_n varies with direction. Its extreme values are found by solving the eigenvalue problem for the **shape operator** (Weingarten map) $\mathcal{S}$, defined by $\mathcal{S}(\mathbf{t}) = -d\hat{\mathbf{n}}(\mathbf{t})$.

In matrix form relative to the basis $\{\mathbf{r}_u, \mathbf{r}_v\}$, the condition $\kappa_n = \lambda$ gives the characteristic equation:

$$\det \begin{pmatrix} L - \lambda E & M - \lambda F \\ M - \lambda F & N - \lambda G \end{pmatrix} = 0,$$

whose roots $\kappa_1 \leq \kappa_2$ are the **principal curvatures**. The corresponding directions are the **principal directions**.

3.2 Gaussian and Mean Curvature

Definition 4. The **Gaussian curvature** and **mean curvature** are

$$K = \kappa_1 \kappa_2 = \frac{LN - M^2}{EG - F^2}, \quad H = \frac{\kappa_1 + \kappa_2}{2} = \frac{EN - 2FM + GL}{2(EG - F^2)}.$$

Remark. By Gauss's *Theorema Egregium*, K is an *intrinsic* invariant: it depends only on E, F, G and their derivatives, not on L, M, N directly.

4 Geometric Interpretation

Geometric Interpretation. The second fundamental form and normal curvature have the following geometric meanings.

(i) Signed distance to the tangent plane. Let P be a point on the surface with tangent plane Π_P . For a nearby point $Q = \mathbf{r}(u_0 + \delta u, v_0 + \delta v)$, Taylor expansion gives

$$\overrightarrow{PQ} = \underbrace{(\mathbf{r}_u \delta u + \mathbf{r}_v \delta v)}_{\in \Pi_P} + \frac{1}{2}(\mathbf{r}_{uu} \delta u^2 + 2\mathbf{r}_{uv} \delta u \delta v + \mathbf{r}_{vv} \delta v^2) + O(\delta^3).$$

The signed distance from Q to the tangent plane is

$$d(Q, \Pi_P) = \overrightarrow{PQ} \cdot \hat{\mathbf{n}} = \frac{1}{2}(L \delta u^2 + 2M \delta u \delta v + N \delta v^2) + O(\delta^3) = \frac{1}{2} \Pi(\delta u, \delta v) + O(\delta^3).$$

Therefore, *the second fundamental form measures, to second order, how much the surface deviates from its tangent plane.*

(ii) Curvature of curves on the surface. For any curve γ on the surface, its curvature vector decomposes as

$$\ddot{\gamma} = \kappa_n \hat{\mathbf{n}} + \kappa_g (\hat{\mathbf{n}} \times \dot{\gamma}),$$

where κ_g is the **geodesic curvature** (tangential bending) and κ_n is the normal curvature. All curves through P in the same tangent direction share the same κ_n (this is *Meusnier's theorem*).

(iii) Sign of κ_n and surface shape. At a point P :

- $K > 0$ (κ_1 and κ_2 same sign): surface is *elliptic* (bowl-shaped); the surface lies on one side of the tangent plane nearby.
- $K < 0$ (κ_1 and κ_2 opposite signs): surface is *hyperbolic* (saddle-shaped); the surface crosses the tangent plane in every neighbourhood.
- $K = 0$ (at least one principal curvature zero): surface is *parabolic* (e.g. a cylinder) or *flat* (a plane).

(iv) **Euler's formula for normal curvature.** If the coordinate directions are chosen to coincide with the principal directions, then $M = 0$ (so $F = 0$ and $E = G = 1$ for a unit-speed frame), and for a tangent direction making angle θ with the first principal direction,

$$\kappa_n(\theta) = \kappa_1 \cos^2 \theta + \kappa_2 \sin^2 \theta.$$

This is **Euler's formula**: normal curvature varies smoothly between the two extreme values κ_1 and κ_2 .